

Michael Deng

AI Operations Portfolio

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This portfolio showcases three production AI-powered automation systems I designed, built, and deployed at Lucid Motors (Sep 2022 – Present). Each workflow runs in production daily, processing thousands of compliance-critical transactions monthly with measurable business impact. All systems were built end-to-end using Python + Playwright with AI-assisted development (Claude, Cursor) and deployed org-wide via one-line push. I own the full lifecycle: problem identification → system design → production deployment → monitoring → on-call.

ARTIFACT 1: COMPLIANCE AUTOMATION BOT (CSN Bot)

Problem

Financial Services Ops manually processed 4,000+ credit score disclosure notices per month. Each required 8 discrete steps across 3 systems (Salesforce → Dealertrack → customer email), taking ~4.3 minutes per transaction. At scale, this consumed 65–70+ hours/week of specialist time with constant risk of compliance errors on federally mandated disclosures.

Solution

End-to-end Python + Playwright headless browser automation deployed org-wide via one-line push:

```
SFDC Report (filtered queue)
→ Open Delivery Object → Match state to dealership
→ Navigate to Dealertrack Unifi Portal
→ Enter Credit Bureau module → Pull Experian report
→ Enter Compliance module → Generate Risk-Based Pricing Notice
→ Save PDF locally as 'Credit Score Notice'
→ Return to SFDC → Open Finance Object → Attach PDF
→ Send templated email to customer on file
→ Tag #CSNS in Sales Memo field ✓
```

Stood up standardized VS Code environments for 3 teammates and 2 adjacent teams (Risk & Compliance, Lease Funding). Provided hands-on coaching for cross-functional RPA adoption.

Impact

Transactions Processed	4,000+/month at 100% accuracy
Manual Hours Eliminated	200+ hours/month (~\$60K+ annual cost savings)
Compliance Errors	Zero in 18+ months of production
Peak Volume	4,287 transactions in a single month, single bot instance
Scale	1 operator → 5 users across 3 teams

Tech Stack: Python · Playwright · Headless Chromium · Salesforce API · Claude (code generation + iteration) · VS Code

ARTIFACT 2: FRAUD DETECTION BOT (LexisNexis InstantID + Emailage)

Problem

Risk & Compliance needed automated identity verification and fraud risk scoring on every financing application. Manual checks were inconsistent, unscalable, and created audit gaps. The Global Head of Risk & Compliance required a solution with configurable scan frequency, one-click ad-hoc capability, and a full audit trail for regulatory examination.

Solution

Python + Playwright bot integrating two LexisNexis risk intelligence APIs, designed in collaboration with Risk & Compliance leadership:

Salesforce (flagged applications queue)
→ LexisNexis InstantID API → Identity verification score
→ LexisNexis Emailage API → Email/phone fraud risk score
→ Pass/Fail determination (configurable thresholds)
→ Output file → SharePoint folder (audit-ready)
→ 100% of FAILS → routed to manual R&C; case review
→ PASSES → sampled for QA (volume-dependent)

Features: Configurable scan frequency (batch or continuous)
One-click ad-hoc scan for manual spot checks
Full SharePoint audit trail for regulatory readiness

Impact

Coverage	100% of flagged transactions processed automatically
Bottleneck Removed	Eliminated manual identity verification for R&C;
Audit Readiness	Full trail stored in SharePoint from day one
Scalability	Enabled fraud screening to scale without headcount
Follow-on	Scoped FICO score automation as next initiative with R&C;

Tech Stack: Python · Playwright · LexisNexis InstantID API · LexisNexis Emailage API · SharePoint · Salesforce · Claude (development acceleration)

ARTIFACT 3: AUTOMATED FUNDING SUBMISSION BOT (S4F)

Problem

After vehicle delivery, specialists manually assembled funding packages (invoice, title application, delivery acceptance, stipulations) across Salesforce and Dealertrack, then submitted to lender. Each submission required 15+ clicks across 2 systems, taking ~12 minutes per deal. At 600+ monthly deliveries, this consumed 28–32 hours/week.

Solution

Event-driven Python + Playwright bot triggered by Salesforce delivery status change:

Salesforce Delivery Status = "Confirmed for Delivery"
→ Bot triggers automatically
→ Pulls funding docs from SFDC (invoice, title app, stips)
→ Navigates to Dealertrack Contract tab
→ Uploads each document with correct doc type mapping
→ Selects all documents → Submits to lender
→ SFDC stage updated to "Funding Pending w/ Lender" ✓

Impact: Before → After

Metric	Before	After
Time per submission	~12 minutes	~90 seconds
Weekly hours consumed	28–32 hours	Under 3 hours
Automation rate	0%	~90%
Error rate	Manual entry errors	Near-zero
EOQ handling	Bottleneck at quarter-end	Scaled seamlessly

Tech Stack: Python · Playwright · Salesforce API · Dealertrack API · Claude (code scaffolding + iteration)

All three systems are live in production. Combined, they process 5,000+ transactions/month, save 230+ manual hours/month, and have maintained zero compliance errors across 18+ months. Each was designed, built, and deployed end-to-end by me — from problem identification through production monitoring and on-call support.

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